Sahaj Bhadja

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Summary

Software Engineer with 2+ years building **infrastructure automation**, monitoring tools, diagnostic workflows, and scalable web applications using Python, React, TypeScript, Ansible, Linux, shell scripting, Kubernetes, Docker, REST APIs, and AI-assisted development.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**

Master of Science in Computer Science *GPA: 3.75/4*

Coursework: Web Technologies, Database Systems, Analysis of Algorithms, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**

Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*

Coursework: Software Engineering, Object-Oriented Programming, Operating Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**

Software Engineer *Los Angeles, CA*

- Built a **multi-tenant operations platform** with React, FastAPI, Azure SQL, and indexed monitoring dashboards, reducing query latency from 1.2 seconds to 350 milliseconds for 500+ daily users.
- Reworked asynchronous workers with explicit job states, failure isolation, and **diagnostic logging**, tripling throughput for 1K+ concurrent jobs while sustaining 99%+ reliability across 5K+ monthly production runs.

Grad8 **June 2025 – August 2025**

Software Engineer Intern *Los Angeles, CA*

- Developed **React and Flask** interfaces for uploads, processing status, and troubleshooting feedback, supporting 150+ instructors and 10K+ daily requests through responsive dashboards and tested production REST APIs.
- Automated concurrent ingestion for 1K+ files per batch using ThreadPoolExecutor, AWS RDS, and **CI/CD validation**, reducing peak processing turnaround 60% while improving production workflow reliability across demanding workloads.

USC AD Labs **January 2025 – May 2025**

ML Research Assistant *Los Angeles, CA*

- Built a Python **benchmarking pipeline** for autonomous-driving models, processing 100K+ trajectory logs through reproducible workloads and improving simulated planning performance 18% across controlled evaluation scenarios and system configurations.
- Profiled spatial-temporal model workloads across latency, accuracy, and failure modes, improving closed-loop stability 15% while documenting **performance analysis** procedures and reproducible results for engineering collaborators across daily workloads.

Aasma Technology Solutions **August 2023 – July 2024**

Software Developer Engineer – SecureSync *India*

- Built cross-platform **automation and integration services** using Python, REST APIs, AWS Glue, and S3, improving pipeline efficiency 68%, reducing response times 30%, and simplifying daily operational maintenance.
- Maintained 20+ unit and integration tests through **Jenkins automation**, reliably diagnosing failures across Windows, Linux, and macOS while reducing technical debt 20% through modular object-oriented service design.

Aasma Technology Solutions **May 2023 – August 2023**

Full Stack Developer Intern – VibrantChem *India*

- Built a **React and Node.js** management application with REST APIs, WebSockets, permissions, and monitoring, synchronizing 2M+ operational records across locations while reducing staged-release conflicts 33% during production deployments.
- Owned eight services through **Docker and Jenkins** deployment, adding automated validation, alerts, and troubleshooting workflows that cut release time 50% while sustaining 95%+ reliability across production environments.

Skills

Languages: Python, TypeScript, JavaScript, Bash, Shell Scripting, C++, Java, SQL
Frontend: React, Redux Toolkit, Next.js, HTML5, CSS3, WebSockets, Responsive Design, Dashboards
Infrastructure: Ansible, Linux, Unix, Kubernetes, Docker, Redfish, IPMI, Provisioning, Configuration Management
Backend & Data: FastAPI, Flask, Node.js, REST APIs, PostgreSQL, SQL Server, MongoDB, Redis, AWS, Azure
Operations & AI: Git, CI/CD, Jenkins, PyTest, Monitoring, Diagnostics, Troubleshooting, Benchmarking, Performance Analysis, LangChain, CrewAI, LLMs

Projects

NodeFleet Control Plane | *Python, React, Redux Toolkit, Ansible, Redfish, Kubernetes*

- Built a **data-center management console** that discovers Redfish nodes, provisions operating systems, applies BIOS and BMC configurations through Ansible, and reliably monitors health across 1,000 simulated servers.

BenchGuard Diagnostics | *Python, Bash, Linux, Prometheus, Docker, LangChain*

- Built an **automated diagnostic toolkit** that runs CPU, memory, storage, and network benchmarks, detects regressions from Prometheus metrics, and uses a LangChain agent to summarize probable root causes.

CPU-Scheduling Simulator | *C++, React, JavaScript, Docker*

- Implemented six **CPU scheduling algorithms** in C++ with interactive React controls, validating 1K+ process combinations and clearly visualizing utilization, waiting time, and throughput to compare system-performance tradeoffs.